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CARBON DIOXIDE ELECTROLYSIS OPERATION MODE

Abstract

A CO.sub.2RR and a CO.sub.2RR electrochemical cell (**2**) is provided, comprising a three-phase boundary (**14**) having: a gaseous CO.sub.2 diffused at least partially through a porous cathode (**8**), a solid catalyst (**10**) operatively associated with the cathode **8,** and a catholyte (**12**) or membrane (**16**) in communication with the catalyst (**10**); a pressure controller (**30**) adapted to control back pressure within, at or near the three-phase boundary (**14**); and a reaction product of the CO.sub.2RR selected from the group consisting of: hydrocarbon, alcohol, H.sub.2 and CO.

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Background/Summary

BACKGROUND
1. Field
[0001] The present invention relates in general to a carbon dioxide (CO.sub.2) electrolysis operation mode, and more specifically to an improved CO.sub.2 reduction reaction (CO.sub.2RR) utilizing back pressure, and most specifically to a CO.sub.2RR electrochemical cell operation mode utilizing back pressure.
2. Description of the Related Art
[0002] CO.sub.2RR has many beneficial aspects. One aspect involves the clean generation of reaction products, such as hydrocarbons, alcohols, H.sub.2 and/or CO, that then can be used to feed any of a variety of industrial, commercial, individual or common inputs such as factories, gas turbines, equipment, vehicles, etc., as well as repositories and other storage mediums, as well as ambient or polluted air. Another aspect involves the consumption or destruction of greenhouse gas CO.sub.2, such as from CO.sub.2 emitting industrial, commercial, individual or common sources such as factories, gas turbines, equipment, vehicles, etc., as well as repositories and other storage mediums, as well as ambient or polluted air. Yet another aspect involves integration of CO.sub.2RR with renewable energy sources such as solar, wind, geothermal, etc., whereby the renewable energy sources provide the requisite electricity or E.sub.o to initiate or sustain the CO.sub.2RR. Still another aspect involves combinations of two or more the above aspects, such as consuming CO.sub.2 flue gas from a factory source while sustaining the CO.sub.2RR from electricity generated from solar panels on the factory roof and then storing or feeding the produced value-added reaction product, such as ethene or higher order hydrocarbons, for a desired use.
[0003] Current CO.sub.2RR, electrochemical cells and operation modes, however, each suffer from one or more of a variety of problems, including difficulty of achieving product selectivity due to the narrow thermodynamic equilibrium potential (E.sub.o) range between various reaction products such as:

CO.sub.2+H.sub.2O+2*e*.sup.−.fwdarw.HCOO.sup.−+OH.sup.− at E.sub.o [V v. SHE] of: −0.12

CO.sub.2+H.sub.2O+2*e*.sup.−.fwdarw.CO+OH.sup.− at E.sub.o [V v. SHE] of: −0.10

2H.sup.++2*e*.sup.−.fwdarw.H.sub.2 at E.sub.o [V v. SHE] of: 0

CO.sub.2+5H.sub.2O+*e*−.fwdarw.CH.sub.3OH+6OH.sup.− at E.sub.o [V v. SHE] of 0.02

2CO.sub.2+8HO+12*e*.sup.−.fwdarw.C.sub.2H.sub.4+12OH.sup.− at E.sub.o [V v. SHE] of: 0.08

2CO.sub.2+9H.sub.2O+12*e*.sup.−.fwdarw.C.sub.2H.sub.5OH+12OH.sup.− at E.sub.o [V v. SHE] of: 0.09

3CO.sub.2+13H.sub.2O+18*e*.sup.−.fwdarw.C.sub.3H.sub.7OH+18OH.sup.− at E.sub.o [V v. SHE] of: 0.10

2CO.sub.2+6H.sub.2O+8*e*.sup.−.fwdarw.CH.sub.3COOH+8OH.sup.− at E.sub.o [V v. SHE] of 0.11

CO.sub.2+6H.sub.2O+8*e*.sup.−.fwdarw.CH.sub.4+8OH.sup.− at E.sub.o [V v. SHE] of: 0.17
[0004] Other CO.sub.2RR, electrochemical cell and operation mode problems may include CO.sub.2 reaction inefficiency and instability over time. Some efforts to overcome these problems involve modifying the catalyst surface such that the CO.sub.2 and its intermediary reactants are more strongly bounded on the catalyst. Other efforts to overcome these problems involve modifying the cathode to optimize gas diffusivity, porosity, thickness, hydrophilicity, electrical conductivity, ionic conductivity, wettability, etc. However, there remains a need for an improved CO.sub.2RR, electrochemical cell and/or operation mode and a particular need for an easily implemented, inexpensive and industrial scalable improved CO.sub.2RR, electrochemical cell, and/or operation mode.

SUMMARY

[0005] In an aspect of the invention, a CO.sub.2RR 2 is provided, comprising a three-phase boundary **14** having: a gaseous CO.sub.2 diffused at least partially through a porous cathode **8**, a solid catalyst **10** operatively associated with the cathode **8**, and a catholyte **12** or membrane **16** in communication with the catalyst **10**; a pressure controller **30** adapted to control the pressure in gas chamber **6** and consequently the back pressure (pressure difference between the gas feed chamber **6** and the catholyte chamber **12**) within, at or near the three-phase boundary **14**; and a reaction product of the CO.sub.2RR such as but not limited to hydrocarbons, alcohols, H.sub.2 and/or CO.

[0006] In another aspect of the invention, an electrochemical cell **2** is provided, comprising: an inlet **4** through which gaseous CO.sub.2 enters the electrochemical cell **2** and advances into a chamber **6**; a porous cathode **8** through which the gaseous CO.sub.2 in the chamber **6** can at least partially diffuse through; a solid catalyst **10** adhered to the cathode **8**; a liquid catholyte **12** in fluid communication with the solid catalyst **10**, wherein the gaseous CO.sub.2 contained in the porous cathode **8** and the solid catalyst **10** and the liquid catholyte **12** collectively form a three-phase boundary **14**; a membrane **16** that separates the cathode **8** from an anode **20** to prevent short circuiting of the electrochemical cell **2** while allowing cations to circulate between the liquid catholyte **12** and a liquid anolyte **18**; an outlet **22** through which reaction product(s) of the CO.sub.2RR exits the electrochemical cell **2**; a pressure sensor **24** arranged within, at or near the three-phase boundary **14** to detect the pressure in catholyte chamber **12** within, at or near the three phase boundary **14**; and a pressure controller **30** adapted to adjust the pressure in gas chamber **6** and consequently the back pressure (pressure difference between the gas feed chamber **6** and the catholyte chamber **12**) within, at or near the three-phase boundary **14**. The pressure controller **30** maintains a pressure difference of 0-400 mbar and preferably 30-130 mbar between gas chamber **6** and catholyte chamber **12**.

[0007] In another aspect of the invention, a, CO.sub.2RR 2 is provided, comprising a three-phase boundary **14** having: a gaseous CO.sub.2 diffused at least partially through a porous cathode **8**, a solid catalyst **10** operatively associated with the cathode **8**, and a liquid catholyte **12** or membrane **16** in communication with the catalyst **10**; a pressure controller **30** that controls the pressure within the three-phase boundary **14** in order to increase a dwell time of the CO.sub.2RR 2 within the three-phase boundary **14**; and a reaction product of the CO.sub.2RR such as but not limited to hydrocarbons, alcohols, H.sub.2 and/or CO.

[0008] In another aspect of the invention, an electrochemical cell **2** is provided, comprising: an inlet **4** through which gaseous CO.sub.2 enters the electrochemical cell **2** and advances into a chamber **6**; a porous cathode **8** through which the gaseous CO.sub.2 in the chamber **6** can at least partially diffuse through; a solid catalyst **10** adhered to the cathode **8**; a liquid catholyte **12** in fluid

communication with the solid catalyst **10**, wherein the gaseous CO.sub.2 contained in the porous cathode **8** and the solid catalyst **10** and the liquid catholyte **12** collectively form a three-phase boundary **14**; a membrane **16** that separates the cathode **8** from an anode **20** to prevent short circuiting of the electrochemical cell **2** while allowing cations to circulate between the liquid catholyte **12** and a liquid anolyte **18**; an outlet **22** through which reaction product(s) of the CO.sub.2RR exits the electrochemical cell **2**; pressure sensors **24**, **26** arranged near the three-phase boundary **14** to detect the pressure within the three phase boundary **14**; and a pressure controller **30** adapted to adjust the back pressure detected by the pressure sensors **24**, **26** the pressure in gas chamber **6** and consequently the back pressure between 0-400 mbar in order to increase a dwell time of the CO.sub.2RR within the three-phase boundary **14**.
[0009] These and other features, aspects and advantages of the present invention will become better understood with reference to the following drawings, description and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] The invention is shown in more detail by help of figures. The figures show preferred configurations and do not limit the scope of the invention.
[0011] FIG. **1** is a detail view of a CO.sub.2RR three phase boundary utilizing back pressure in accordance with an exemplary embodiment of the subject matter.
[0012] FIG. **2** is a schematic of a CO.sub.2RR electrochemical cell operation mode utilizing back pressure in accordance with an exemplary embodiment of the subject matter.
DETAILED DESCRIPTION
[0013] In the following detailed description of the present invention, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration, and not by way of limitation, specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized and that changes may be made without departing from the spirit and scope of the subject matter or present invention.
[0014] As illustrated in FIG. **1**, a CO.sub.2RR utilizing back pressure to control CO.sub.2 dwell or occupation within, at or near a three-phase boundary and thereby control the CO.sub.2RR and desired reaction product(s) is provided. The illustrated three-phase boundary comprises a gaseous CO.sub.2 diffused at least partially through a porous cathode **8**, a solid catalyst **10** operatively associated with the porous cathode **8**, and a liquid (or solid polymer) catholyte **12** or membrane **16** in communication with the catalyst **10**. Establishing a back pressure, at or near the three-phase boundary **14** to 0-400 mbar and preferably to 30-130 mbar is particularly suited for the efficient generation of hydrocarbon product(s). Reaction products may include hydrocarbons (of either or both higher and lower order), alcohols, H.sub.2 and/or CO. If no catholyte **12** is used, the catholyte **12** alkaline environment functionality may be achieved by membrane **16**. Also in this catholyte-less configuration, the three-phase boundary **14** becomes the cathode **8**, catalyst **10** and membrane **16**, with membrane **16** liquidity source-able from the liquid anolyte **18** or other suitable liquid source associated with the cell **2**.
[0015] As illustrated in FIG. **2**, an operation mode utilizing back pressure on a CO.sub.2RR electrochemical cell **2** is also provided. The illustrated electrochemical cell **2** comprises a CO.sub.2 inlet **4**, a three-phase boundary **14** comprising a porous cathode **8** through which gaseous CO.sub.2 is diffused, a solid catalyst **10** adhered to the cathode **8**, a liquid catholyte **12** in fluid communication with the catalyst **10**, as well as a membrane **16** that separates the cathode **8** from an anode **20** **2** while allowing cations to circulate between the liquid catholyte **12** and a liquid analyte **18**, along with a reaction product outlet **22**. A pressure sensor **24** is arranged within, at or near the three-phase boundary **14** to detect the pressure in catholyte chamber **12** within, at or near the three-

phase boundary **14**. A pressure controller **30** is adapted to control the pressure in gas chamber **6** and consequently the pressure difference (back pressure) between gas chamber **6** and catholyte chamber **12**. The pressure controller **30** maintains a pressure difference of 0-400 mbar and preferably 30-130 mbar between gas chamber **6** and catholyte chamber **12**. However, as will be understood by those skilled in the art, depending on the desired CO.sub.2RR reaction product, particular electrochemical cell **2** configuration, three-phase boundary **14** constituents, and a variety of other variable factors such as gas diffusion layer constitution and operation conditions including applied current density, temperature, flow rates and the like, not all of the elements shown in FIG. **2** are necessary to provide a CO.sub.2RR electrochemical cell **2** operation mode utilizing back pressure in order to generate CO.sub.2RR products.

[0016] Referring still to FIG. **2**, the electrochemical cell **2** has a CO.sub.2 inlet **4** configured to receive CO.sub.2 gas from any one or more of a variety of CO.sub.2 sources, including without limitation, industrial, commercial, individual or common sources such as factories, gas turbines, equipment, vehicles, etc., as well as repositories and other storage mediums, as well as ambient or polluted air. Depending on the desired reaction product, electrochemical cell **2** configuration, selected three-phase boundary **14** constituents, or a variety of other factors, the CO.sub.2 optionally may be conditioned before, within or after the inlet **4**, such as by humidification via H.sub.2O or other suitable means. FIG. **1** exemplarily illustrates humidified CO.sub.2 passing through the inlet **4** into a chamber **6** that holds and allows dispersion of the CO.sub.2. Of course, no or more than one inlet **4** and no or more than one chamber **12** can be used, and any inlet(s) **4** and chamber(s) **6** used can be arranged at different location(s) on, along, or through the electrochemical cell **2**.

[0017] A porous cathode **8** is arranged to receive the gaseous CO.sub.2 and configured such that the CO.sub.2 can diffuse through at least a portion of the cathode **8**. The exemplarily illustrated cathode **8** is a gas diffusion electrode (GDE) that absorbs and converts the CO.sub.2 molecules into the desired CO.sub.2RR reaction product e.g. hydrocarbons, however other suitable cathodes **8** could be used. A solid catalyst **10** is operatively associated with the cathode **8** by any suitable means, such as drop casting, plating, doping, or spray coating to enhance reaction product selectivity as well as reaction efficiency and stability. Suitable catalysts **10** include but are not limited to Pt, Zn, Cu, Ag, Au, Pd and Sn. Since Cu is the only transition metal catalyst for CO.sub.2RR to value added C.sub.2+ reaction products e.g. ethene, ethanol, propanol, Cu is therefore preferred but not required when desiring those reaction products.

[0018] A catholyte **12** is advantageously arranged in communication with the catalyst **10** to provide an alkaline environment close to the three-phase boundary **14** and thereby promoting CO.sub.2RR thermodynamically and kinetically when hydrocarbon reaction product(s) and/or a CO reaction product is desired. The exemplarily illustrated catholyte **12** is an alkaline buffer liquid solution, such as potassium hydrogen carbonate, cesium hydrogen carbonate, rubidium hydrogen carbonate, lithium bicarbonate or potassium hydroxide, but the catholyte **12** may also be embodied as a solid polymer.

[0019] A three-phase boundary **14** is thereby formed by: (1) the gaseous CO.sub.2 diffused at least partially through the cathode **8**, (2) the solid catalyst **10** operatively associated with the cathode **8**, and (3) the liquid (or solid polymer) catholyte **12** in communication with the catalyst **10**. FIG. **1** provides a detailed illustration of the three-phase boundary, exemplary H+ and CO.sub.2 reaction constituents, as well as exemplary desired and undesired CO.sub.2RR reaction products.

[0020] Depending on the desired reaction product(s), electrochemical cell **2** configuration, selected three-phase boundary **14** constituents, or a variety of other variable factors such as gas diffusion layer constitution and operation conditions including applied current density, temperature, flow rates and the like, optionally, a membrane **16** optionally may be used to separate the cathode **8** from an anode **20** while allowing cations to circulate between the liquid catholyte **12** and a liquid anolyte **18**. Exemplary suitable membranes **16** include sulfonated tetrafluoroethylene based fluoropolymer-copolymer (Nafion). If used, the anolyte **18** is advantageously an alkaline buffer solution to complete the ion circuitry of the electrochemical cell **2**. Exemplary suitable analytes **18** include H_2O, OH—, H^+ and electrolyte associated cations and anions (e.g. Cs^- and SO_4^{-2}). Also if used, the anode **60** material can be composed of Ir, Ni and/or Pt, where the anode would be responsible for O_2 evolving reaction and completing the ion circuit in the electrochemical cell by pumping H^+ to the cathode.

[0021] Similarly depending on the desired reaction product(s), electrochemical cell **2** configuration, selected three-phase boundary **14** constituents, or a variety of other variable factors such as gas diffusion layer constitution and operation conditions including applied current density, temperature, flow rates and the like, optionally, the membrane **16**, preferably an anion exchange membrane **16** or a bipolar membrane **16**, may be used to sustain the alkaline environment rather than or in addition to the catholyte **12**. In this configuration, no catholyte **12** is used and the catholyte **12** alkaline environment functionality is achieved by membrane **16**. Also in this catholyte-less configuration, the three-phase boundary **14** becomes the cathode **8**, catalyst **10** and membrane **16**, with membrane **16** liquidity source-able from the liquid anolyte **18** or other suitable liquid source associated with the cell **2**.

[0022] A reaction product outlet **22** is configured to receive the reaction product, e.g. hydrocarbon gas, from the electrochemical cell **2** and feed the reaction product to any one or more use sources, including without limitation, industrial, commercial, individual or common sources such as factories, gas turbines, equipment, vehicles, etc., as well as repositories and other storage mediums, as well as ambient or polluted air. Depending on the desired reaction product, electrochemical cell configuration **2**, selected three-phase boundary **14** constituents, or a variety of other factors such as reaction product use source(s), the CO_2 reaction product optionally may be conditioned before, within or after the outlet **22**, such as by liquifying (e.g. to form LNG particularly if methane is a reaction product). Unconditioned reaction products are exemplary shown as passing through the outlet **22**. Of course, no or more than one outlet **22** can be used, and any outlet(s) **22** used can be arranged at different location(s) on, along, or through the electrochemical cell **2**.

[0023] Still referring to FIG. **2**, a pressure sensor **24** is arranged in operative communication with the electrochemical cell **2** in order to detect pressure in catholyte chamber **12** within, at or near the three-phase boundary **14**. The exemplary embodiment shows the pressure sensor **24** arranged near the three-phase boundary **14**, toward the middle of the catholyte **12** between the cathode **8** and membrane **16**, however, depending on the desired reaction product(s), electrochemical cell **2** configuration, three-phase boundary **14** constituents, or a variety of other factors, the pressure sensor **24** may be arranged in any of a variety of locations within, at or near the three-phase boundary **14**. Also, if the three-phase boundary **14** is expansive or geometrically complex, then pressure sensors **26**, **28** advantageously may be used to better sense the three-phase boundary **14** pressure and control the CO_2RR.

[0024] An outlet pressure sensor **26** may be advantageously arranged within, at or near the outlet **22** in order to detect pressure within, at or near the outlet **22**. The exemplary illustration shows an outlet pressure sensor **26** arranged at the beginning of the outlet **22**. Of course, depending on the outlet **22** configuration, one or more optional outlet pressure sensors **26** could be arranged in any of a variety of locations within, at or near the outlet **22** to better sense the outlet pressure and control the CO_2RR. Also optionally, an inlet pressure sensor **28** may be arranged within, at or near the inlet **4** in order to detect CO_2 pressure within, at or near the inlet **4**. The exemplary illustration shows an optional inlet pressure sensor **28** arranged at the end of the inlet **4**. Of course, depending on the inlet configuration, one or more optional outlet pressure sensors **28** could be arranged in any of a variety of locations within, at or near the inlet **4** to better sense the inlet pressure and control the CO_2RR.

[0025] A pressure controller **30** is arranged in operative communication with the pressure sensor(s) **24**, **26** and/or optional pressure sensor **28** and is adapted to adjust pressure detected by the pressure sensor(s) **24**, **26** and/or optional pressure sensor **28** in order to adjust the back pressure within, at or near the three phase boundary **14** and/or outlet **22** or inlet **4**. The exemplary illustration shows the pressure controller **30** located downstream of the outlet **22** and outlet pressure sensor **26** and embodied as a membrane valve with a flexible diaphragm that allows outlet flow only at, above, below or between a desired pressure. Of course, depending on the desired reaction product(s), electrochemical cell **2** configuration, three-phase boundary **14** constituents, and a variety of other factors such as desired pressure(s), etc., the pressure controller **30** may be embodied through any of a variety of suitable mechanisms and may be arranged in any of a variety of locations. Also, if the electrochemical cell **2**, CO.sub.2 feed source, or reaction product feed use is expansive or complex, then additional pressure sensors **30** may be advantageously used to better control and control the CO.sub.2RR.

[0026] Still referring to FIG. **2**, an exemplary illustration of CO.sub.2RR electrochemical cell **2** operation mode utilizing back pressure is provided. Pressurized humidified gaseous CO.sub.2 enters electrochemical cell **2** through gas inlet **4** and into chamber **6** in the range of 95-10 vol %, preferably 80-30 vol %, in ΔP=0-400 mbar, preferably 50-200 mbar and V.sub.CO2=5-200 sccm, preferably 15-50 sccm, The CO.sub.2 is urged toward cathode **8** and diffuses through at least a portion of cathode **8**, thereby contacting catalyst **10**. The CO.sub.2RR thereby occurs within, at or near the three-phase boundary **14**, with the reaction product then advancing through the outlet **22** for desired collection or use.

[0027] Reaction product selectivity, as well as efficiency and stability, is improved by controlling the CO.sub.2 pressure within, at or near the three-phase boundary **14**. One way to control the CO.sub.2 pressure within, at or near the three-phase boundary **14** is via the pressure controller **30** arranged in operative communication with the pressure sensor(s) **24** and **26**, and/or optional pressure sensor **28** to thereby adjust pressures detected by the pressure sensor(s) **24** and **26**, and/or optional pressure sensor **28** in order to adjust the back pressure within, at or near the three-phase boundary **14** and/or outlet **22** or inlet **4**.

[0028] In an exemplary CO.sub.2RR operation, CO.sub.2 inlet **4** pressure (P1) is sensed as atmospheric at 1 mbar (within an exemplary preferred range of 0 mbar to 10 mbar), while three-phase boundary **14** pressure (P2) is sensed at 20 mbar (outside an exemplary preferred range of 30 mbar to 130 mbar). Based on the P2 and P1 pressure difference and the preferred pressure ranges, the three-phase boundary **14** pressure (P2) is adjusted to a pressure of 85 mbar (or anywhere within the exemplary preferred range of 30 mbar to 130 mbar) in order to better control the CO.sub.2RR and desired reaction product.

[0029] In a second exemplary operation, CO.sub.2 outlet **22** pressure (P3) is 15 mbar (within an exemplary preferred range of 10 mbar to 30 mbar), while three-phase boundary **14** pressure (P2) varies between 120-150 mbar (partially within and partially outside an exemplary preferred range of 30 mbar to 130 mbar). Based on the P2 and P3 pressure difference and the preferred pressure ranges, the three-phase boundary **14** pressure (P2) is continually adjusted to a desired set pressure of 70 mbar (or anywhere within the exemplary preferred range of 30 mbar and 130 mbar) in order to better control the CO.sub.2RR and desired reaction products.

[0030] In a third exemplary operation, CO.sub.2 inlet **4** pressure (P1) is 2-4 mbar as continually measured by inlet pressure sensor **28**, while three-phase boundary **14** pressure (P2) is 60-75 mbar as continually measures by pressure sensor **24**, and outlet **22** pressure (P3) is 5-10 mbar as continually measured by outlet pressure sensor **26**. Based on the P3, P2 and P1 pressure differences, as well as an exemplarily desired three-phase boundary **14** pressure range of 0-400 mbar (P3), an exemplarily desired inlet **4** pressure of 1 mbar (P1) and an exemplarily desired outlet **22** pressure range of 2-5 mbar (P3), controller **30** periodically adjusts the three-phase boundary **14** pressure (P2) to 55 mbar (or anywhere between 0 mbar and 400 mbar), the inlet **4** pressure (P1) to 1 mbar and the outlet **22** pressure (P3) to 4 mbar, in order to better control the CO.sub.2RR and desired reaction products.

[0031] In a fourth exemplary operation, CO.sub.2 inlet **4** pressure (P1) is 50-75 mbar as continually

measured by inlet pressure sensor **28**, while three-phase boundary **14** pressure (P2) is 150-180 mbar as continually measures by pressure sensor **24**, and outlet **22** pressure (P3) is 250-270 mbar as continually measured by outlet pressure sensor **26**. Based on the P3, P2 and P1 pressure differences, as well as an exemplarily desired three-phase boundary **14** pressure range of 0-400 mbar (P3), an exemplarily desired inlet **4** pressure of 75-100 mbar (P1) and an exemplarily desired outlet **22** pressure range of 300-320 mbar (P3), controller **30** periodically adjusts the three-phase boundary **14** pressure (P2) to 175 mbar (or anywhere between 80-100), the inlet **4** pressure (P1) to 100 mbar and the outlet **22** pressure (P3) to 250 mbar, in order to better control the CO.sub.2RR and desired reaction products.
[0032] In a fifth exemplary operation, a plurality of inlet pressure sensors **28** are employed near inlet **4** from which an average pressure (P1) of 2 mbar is calculated, while a plurality of outlet pressure sensors **26** are employed near outlet **22** from which an average pressure (P3) of 10 mbar is calculated. Based on the P3 and P1 pressure difference, as well as an exemplarily desired three-phase boundary **14** pressure range of 30-130 mbar, an exemplarily desired inlet pressure (P1) of 1 mbar and an exemplarily desired outlet pressure range (P3) of 2-5 mbar, controller **30** adjusts the inlet pressure (P1) to 1 mbar and the outlet pressure (P3) to 3 mbar, in order to better control the CO.sub.2RR and desired reaction products. In this exemplary operation, pressure sensor **24** is not used while pressure sensors **26** and **28** are used.
[0033] Through the above utilization of back pressure with CO.sub.2RR or with a CO.sub.2RR electrochemical cell **2** operation mode, CO.sub.2 availability at the three-phase boundary **14** can be increased. It has been observed that increased CO.sub.2 availability suppresses availability of undesired constituents such as those forming hydrogen evolving reaction (HER) reaction products, enhances the desired CO.sub.2RR product(s), and helps prevent the liquid catholyte **12** from flooding the catalyst **10** or cathode **8**. Thus, the above utilization of back pressure with CO.sub.2RR or with a CO.sub.2RR electrochemical cell **2** operation mode thereby overcomes difficulty of achieving reaction product selectivity as well as CO.sub.2RR reaction inefficiency and instability over time. The need for an improved CO.sub.2RR and the need for an easily implemented, inexpensive and industrial scalable CO.sub.2RR electrochemical cell **2** operation mode is thereby provided through the above selective utilization of back pressure with CO.sub.2RR electrochemical cell **2** operation.
[0034] While specific exemplary embodiments and illustrations have been described in detail, those with ordinary skill in the art will appreciate that various modifications and alternative to those details could be developed in light of the overall teachings of the disclosure. Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of the subject matter, which is to be given the full breadth of the appended claims, and any and all equivalents thereof.

Claims

1. A CO.sub.2RR, comprising: a three-phase boundary (**14**) comprising: a gaseous CO.sub.2 diffused at least partially through a porous cathode (**8**), a solid catalyst (**10**) operatively associated with the cathode (**8**), and a liquid catholyte (**12**) or a membrane (**16**) in communication with the catalyst (**10**); a pressure controller (**30**) that controls pressure within the three-phase boundary (**14**) in order to increase a dwell time of the CO.sub.2RR within the three-phase boundary (**14**); and a reaction product of the CO.sub.2RR formed within the three-phase boundary (**14**) selected from the group consisting of: hydrocarbon, alcohol, H.sub.2 and CO.
2. The CO.sub.2RR of claim 1, further comprising pressure sensors (**24**) and (**26**) arranged within the three-phase boundary (**14**) to detect back pressure within the three-phase boundary (**14**).
3. The CO.sub.2RR of claim 2, wherein the pressure controller (**30**) controls back pressure detected by the pressure sensors (**24**) and (**26**) between 0 mbar-400 mbar.

4. The CO.sub.2RR of claim 3, wherein the cathode (**8**) is a gas diffusion electrode.
5. The CO.sub.2RR of claim 4, wherein a catalyst (**10**) is selected from the group consisting of Cu, Ag, Au, Pd and Sn and the catalyst **10** is adhered to the cathode (**8**) by drop casting, plating, spray-coating or doping.
6. The CO.sub.2RR of claim 5, wherein the catholyte (**12**) is an alkaline buffer liquid solution comprising potassium hydrogen carbonate, cesium hydrogen carbonate, rubidium hydrogen carbonate, lithium bicarbonate or potassium hydroxide, and the liquid catholyte is in fluid communication with the catalyst (**10**).
7. The CO.sub.2RR of claim 5, wherein the catholyte (**12**) is either: (i) an alkaline buffer liquid solution comprising potassium hydrogen carbonate, cesium hydrogen carbonate, rubidium hydrogen carbonate, lithium bicarbonate or potassium hydroxide, or (ii) a solid polymer, and wherein the membrane (**16**) is either an anion exchange membrane or a bipolar membrane.
8. The CO.sub.2RR of claim 7, wherein the reaction product includes ethene.
9. The CO.sub.2RR of claim 8, further comprising an outlet pressure sensor (**26**) arranged within an outlet (**22**).
10. CO.sub.2RR of claim 1, wherein the CO.sub.2RR occurs in an electrochemical cell (**2**) having an inlet (**4**) through which the CO.sub.2 enters the electrochemical cell (**2**) and having an outlet (**22**) through which the reaction product exits the electrochemical cell (**2**).
11. The CO.sub.2RR of claim 10, wherein the electrochemical cell further comprises a membrane (**16**) that separates the cathode (**8**) from an anode (**20**) to prevent short circuiting of the electrochemical cell (**2**) while allowing cations to circulate between the liquid catholyte (**12**) and a liquid anolyte (**18**).
12. A CO.sub.2RR electrochemical cell (**2**), comprising: an inlet (**4**) through which gaseous CO.sub.2 enters the electrochemical cell (**2**) and advances into a chamber (**6**); a porous cathode (**8**) through which the gaseous CO.sub.2 in the chamber (**6**) can at least partially diffuse through; a solid catalyst (**10**) adhered to the cathode (**8**); a liquid catholyte (**12**) in fluid communication with the solid catalyst (**10**), wherein the gaseous CO.sub.2 contained in the porous cathode (**8**) and the solid catalyst (**10**) and the liquid catholyte (**12**) collectively form a three-phase boundary (**14**); a membrane (**16**) that separates the cathode (**8**) from an anode (**20**) to prevent short circuiting of the electrochemical cell (**2**) while allowing cations to circulate between the liquid catholyte (**12**) and a liquid anolyte (**18**); an outlet (**22**) through which a hydrocarbon reaction product of the CO.sub.2RR exits the electrochemical cell (**2**); pressure sensors (**24**) and (**26**) arranged near the three-phase boundary (**14**) to detect back pressure within the three-phase boundary (**14**); and a pressure controller (**30**) adapted to adjust the back pressure detected by the pressure sensors (**24**) and (**26**) between 0 mbar-400 mbar in order to increase a dwell time of the CO.sub.2RR within the three-phase boundary (**14**);
13. The CO.sub.2RR electrochemical cell (**2**) of claim 12 further comprising an inlet pressure sensor (**28**) arranged near the inlet (**4**), and further comprising a second pressure controller (**32**) arranged near the inlet (**4**) upstream the inlet pressure sensor (**28**), and wherein the reaction product is a hydrocarbon, alcohol, H.sub.2 and/or CO.